Mohammed Salim

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Profile

First Class Computer Science graduate (Aston University) and freelance software developer specialising in Python, machine learning, and REST API integration. I have delivered production software for clients, an end-to-end identity-verification API integration for a UK pharmacy and an academic journal publishing platform for the Islamic Cultural Centre (London Central Mosque). My final-year research built complete deep reinforcement learning and computer vision systems in Python from the ground up. I combine strong CS fundamentals with proven, independent delivery, and I am looking to grow rapidly as a Python developer inside a professional engineering team.

Education

Aston University — *Birmingham, UK • 2023 - 2026*

BSc (Hons) Computer Science – First Class Honours

- Final-year project in deep reinforcement learning and computer vision (graded at first-class standard). Core grounding in algorithms, data structures, databases, and software engineering.

Technical Skills

Python (primary): FastAPI, pandas, NumPy, PyTorch, OpenCV, requests; async APIs, data pipelines, scripting & automation

Other languages: SQL, TypeScript, JavaScript, C++, C#, PHP, GoLang

Frameworks & ML: Stable-Baselines3, Gymnasium, OpenCV, NumPy, Matplotlib; Node.js, React, Laravel, Tailwind CSS

Engineering & ML concepts: REST API design & integration, Deep Reinforcement Learning, Proximal Policy Optimisation (PPO), Computer Vision, PyBullet, software architecture,

Databases & tools: PostgreSQL, MySQL, REST APIs; Git & GitHub, Docker, pytest (unit testing), CI, Linux, Supabase

Embedded systems: C/C++, Arduino, ESP32, ESP-NOW, SPI, I2C, UART, GPIO, PWM; Wi-Fi/BLE & MQTT

Projects

AI Video Generation Pipeline — *Python · FastAPI · PyTorch · Docker*

- Built an async, end-to-end media-processing API that turns a single request into a finished video, orchestrating a five-stage pipeline of LLM script generation, TTS, RVC neural voice conversion **(PyTorch)**, video rendering, and cloud delivery.
- Designed a job-queued architecture with per-job status tracking and namespaced artifacts for safe concurrency, and integrated Backblaze B2 **(S3-compatible) object storage** with presigned download URLs.
- Engineered a fail-soft pipeline with per-stage error handling, secured endpoints with API-key auth, containerised with **Docker**, and wrote an isolated **pytest** suite mocking cloud clients so tests run with no models or keys.

Robotic Manipulation via Deep Reinforcement Learning — *Final Year Project*

- Engineered a custom physics simulation environment using **Python** and **PyBullet** to train a Franka Panda robotic arm in dynamic, non-prehensile manipulation tasks.
- Trained a Proximal Policy Optimisation (PPO) policy utilising Stable-Baselines3 and the Gymnasium API, advancing the model through a progressively difficult 5-stage curriculum.

- Built a custom 'Milestone Checkpointing' system during domain randomisation, reaching **76.3% success rate** under ±25 mm spatial noise **and 80.2%** under 0.3 rad arm-pose noise.
- Designed a particle spring-mass system for soft-body simulation, achieving an **81.0% zero-shot generalisation** rate from a rigid-body training phase.
- Integrated a computer vision pipeline using HSV image moments to successfully transition the model from ground-truth data to synthetic camera observations.

Octo Man | IoT Smart Vending Machine — *Aston Hack 11 - 2nd Place Winner*

- Awarded 2nd Place overall at Aston Hack 11, winning four 1440p monitors for engineering a real-time, community-regulated smart dispenser designed to playfully address the "Tragedy of the Commons".
- Developed a Node.js server and an animated Tailwind CSS web interface, implementing bidirectional **WebSockets** to stream live voting data and trigger the physical hardware via parsed JSON payloads using ArduinoJson.
- Interfaced a GC9A01 round LCD over SPI using TFT_eSPI to dynamically render custom bitmap graphics, alongside programming precise PWM rotational and drop mechanisms with ESP32Servo.

Experience

Software Developer (Contract) — *Independent — 2026 - Present*

- Delivered a production identity-verification system end to end for a UK pharmacy (Courier Pharmacy) by integrating the **LexisNexis identity-verification REST API** into their WordPress/WooCommerce site, so ID checks run automatically at customer sign-up with no manual intervention.
- Handled the full workflow — API authentication, request construction, and response validation — and wired the verification flow into WooCommerce, WPForms, and a CRM.
- Packaged the deployment as a self-contained plugin to minimise ongoing maintenance for a non-technical client.

Freelance Web Developer — *London Central Mosque — 2024 - Present*

- Sole developer for the Islamic Cultural Centre (London Central Mosque), the UK's oldest and most prominent Islamic institution (est. 1944), building their academic journal publishing platform modelled on Brill and the Oxford Journal of Islamic Studies.
- Architected and built the current publishing platform in **Laravel/PHP** — user accounts, article indexing and page system, and a Filament-based CMS for editorial staff.
- Built the integration framework for automated DOI registration via the **Crossref API**, designing the request and metadata logic to Crossref's schema for academic database indexing.

Activities

Compete in the Volleyball League as a middle blocker. Ongoing personal projects in Python, applied ML, and electronics (Arduino/ESP32).